

Data Communication.

Unit-I

Introduction:

Data communication: components, presentation, Data Flow. Networks: Network Criteria, Network Topology, Physical structure, Network classification: The Internet, Protocols, and standards. Switching: Message, packet and Circuit Switching

Network Models:

Layered Architecture, The OSI model, TCP/IP Protocol Stack, ARPANET

Unit II

Digital Transmission and Physical Layer

Digital Representation of Information, Analog and Digital Signals, Digital Transmission, Line coding, Modulation, Transmission Modes, Multiplexing: Frequency Division Multiplexing (FDM), Time Division Multiplexing (TDM), Wavelength Division Multiplexing (WDM), SONET

Transmission Media

Guided Media: Twisted Pair, Co-axial cable, Optical Fiber, Unguided - Media.

(Cellular Telephones Generation of Networks: 1G, 2G, 3G, 4G, 5G)

Unit III

Data Link Layer: Data Link Layer Design Issues, ARQ Protocols: STOP and WAIT, GO-BACK 'N', Selective Repeat Protocols. Efficiency of ARQ Protocols, Flow control, Sliding window flow control.

Control Data Link Control : HDLC, PPP, Statistical Multiplexing, Error detection, Parity Bit, Two-dimensional parity checks, Internal checksum, Polynomial codes

Unit IV

Medium Access Control Protocols

Multiple Access Communication Local Area Network LAN structure, MAC sublayer, Logical Link control Layer, Random Access Protocol - ALOHA, Slotted ALOHA, CSMA, CSMA/CD.

Scheduling Approaches to Medium Access

control - Reservation systems Polling,

Token Passing ring, channelization - FDMA,

TDMA, CDMA

LAN Standards:

~~IEEE~~ Ethernet and IEEE 802.3 LAN standard

Token Ring and IEEE 802.5 LAN standard

FDDI, wireless LAN's and IEEE IEEE 802.11

~~IEEE~~ LAN standard